These boards normally require 6 or more volts of drive. The issue is that the optocoupler LED is in series with an LED and a current limiting resistor. The optocoupler voltage drop is about 1.7V at 20 mA. Since the indicator LED is also more than 1.5V, 3.3V drive will not work. The 470 Ohm current limiting resistor would limit drive to the optocoupler to only about 3 mA which is not enough drive for reliable operation.

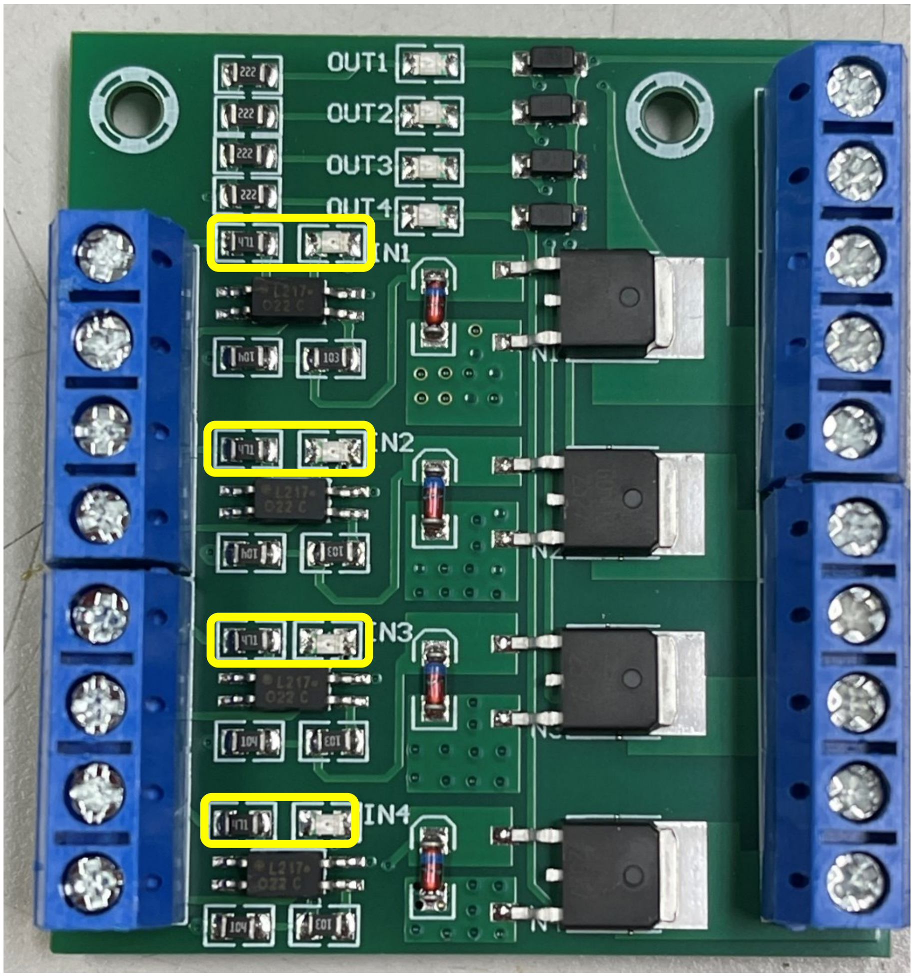

Figure 1 Unmodified YYNMOS-4

The modification involves removing the LED and replacing it with a short, and replacing the current limiting resistor with a lower value. I used 120 ohms which results in about 15 mA of drive – more than enough.

In Figure 2 I used a 120 ohm 0805 size surface mount resistor and a solder blob short. I didn’t have any zero ohm resistors of the correct physical size. I could have used two resistors that added up to the appropriate value, but I had the 120s on hand.

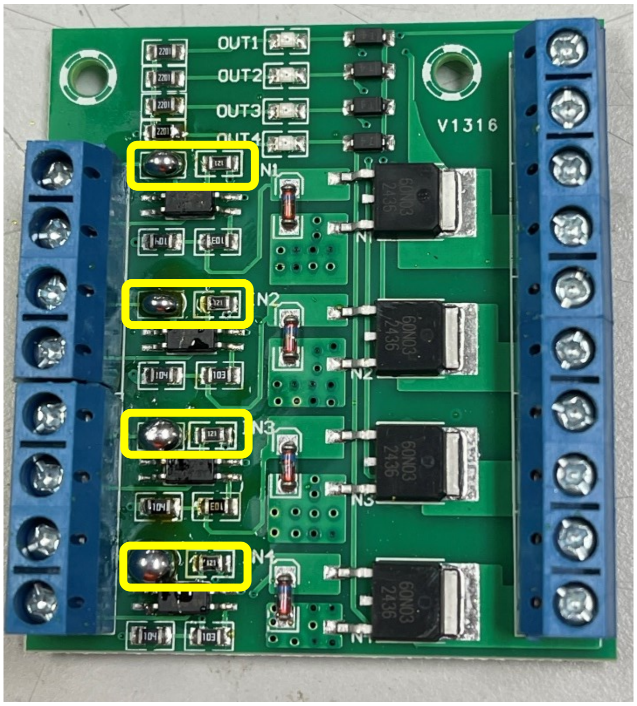

Figure 2 Modified YYNMOS-4